

Session – 1

Task 4 — Why Java is Used for Test Automation (Short Note)

Java is one of the most popular languages for test automation because it is **platform-independent** (Write Once, Run Anywhere), strongly typed, and has a huge community with rich library support. It integrates seamlessly with powerful automation frameworks like **Selenium WebDriver**, which uses Java as its primary language for writing browser automation scripts. Tools like **TestNG**, **JUnit**, and **Maven** are also Java-based, making it a complete ecosystem for end-to-end test automation.

Task 5 — JDK vs JRE vs JVM (In Your Own Words + Analogy)

Term	Full Form	What it does	Real-World Analogy
JVM	Java Virtual Machine	Runs the Java bytecode on your machine. It's the engine.	 Car Engine — you don't see it, but it makes everything run
JRE	Java Runtime Environment	Contains JVM + libraries needed to <i>run</i> Java programs	 Full Car — engine (JVM) + seats, AC, wheels = ready to drive (run apps)
JDK	Java Development Kit	Contains JRE + compiler + tools to <i>write & build</i> Java programs	 Car Factory — you can build a car (JRE) AND drive it too